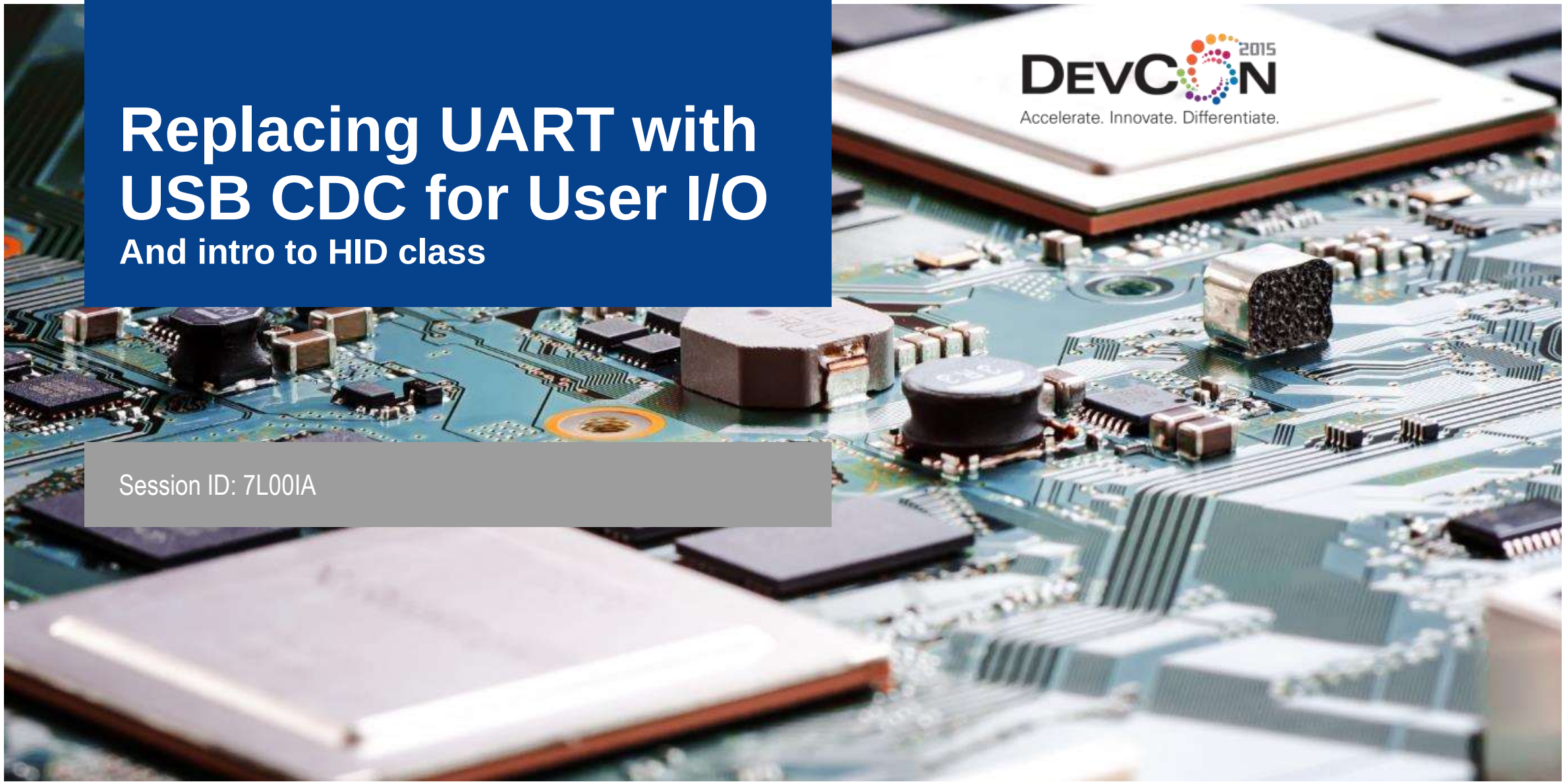
Replacing UART with USB CDC for User I/O

And intro to HID class



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RENESAS



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Replacing UART with USB CDC For User I/O

USB is now everywhere in computing platforms and has mostly replaced RS232 serial UART communications.

- Formerly, connecting an embedded device to a computer-based console often used a serial COM link.
- Now, serial ports are rarely present on today's computers.
- USB is a logical alternative for moving data between a computer and the embedded device.

This lab will help you:

- **Accelerate:** Think of this as a “quick-start guide” to adding a Renesas USB Peripheral class CDC driver code to your embedded project for basic two-way communication with a PC. The emphasis will be on getting data from an embedded device to a PC using USB as an alternative to UART (serial COM) interface. An intro to the HID class will also be explored.
- **Innovate**: Using USB opens the possibility of communicating with a wider range of modern devices.
- **Differentiate:** Renesas USB ‘mini’ FIT drivers let you add USB to smaller embedded projects that others would consider too difficult.

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Agenda

Presentation

- Why Replace UART With USB?
- The challenge of adding USB drivers to embedded projects
- Renesas USB driver FIT modules.
- Which USB class, CDC? HID?
- Overview of CDC class driver
- Overview of HID class driver

Lab

- Step-by-step tutorial exercise: Adding the Renesas USB PCDC driver to an RX project.

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Replacing UART with USB CDC

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Why Replace UART With USB?

The physical RS-232 ‘COM’ port is rapidly disappearing from most computers, especially laptops.

USB is replacing the RS232 serial ‘UART’ port as the de-facto communications interface for PCs, however many computer applications still use the RS-232 standard protocols.

One solution can be to use the USB bus to emulate RS-232. This offers several advantages:

- Every new computer has USB ports.
- The PC application need not be changed.
- Your embedded device can be bus powered.*

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Adding USB drivers to embedded projects can be hard

Why do we avoid adding USB support to our embedded projects?

- Published USB standards documents are complex—hard to understand.
- USB specifications don't focus on how to implement in our environment, or for our specific application.
- What do I really need to understand, and what can I ignore?
- Need to become a specialized USB 'expert' to create my own driver.
- Large time investment.

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Solution: Renesas USB device driver FIT modules.

- **FIT** = Firmware Integration Technology
- USB driver code available as ready-to-use, individual software modules.
  - Peripheral Communications Device Class (PCDC)
  - USB Peripheral Human Interface Device Class (PHID)
- Works with Renesas FIT compliant board support package (BSP)
- Run within a simple ‘task scheduler’ process. (Doesn’t need to be RTOS)
- e2studio plugin to ease addition of module into your project

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Choosing the right USB Class

USB CDC Class

All types of communication services over USB

- Modems, digital telephones
- Networking devices
  - ADSL and cable modems
  - Ethernet devices

- Typical usage: Peripheral and host driver mimic a serial COM-port

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Choosing the right USB Class …

USB HID Class

- The USB human interface device class (USB HID class) specifies a device class for devices such as keyboards, mice, game controllers and alphanumeric display devices.
- Can be adapted to serve as a container class for other data.
- USB ‘Interrupt’ mode = consistent timing.

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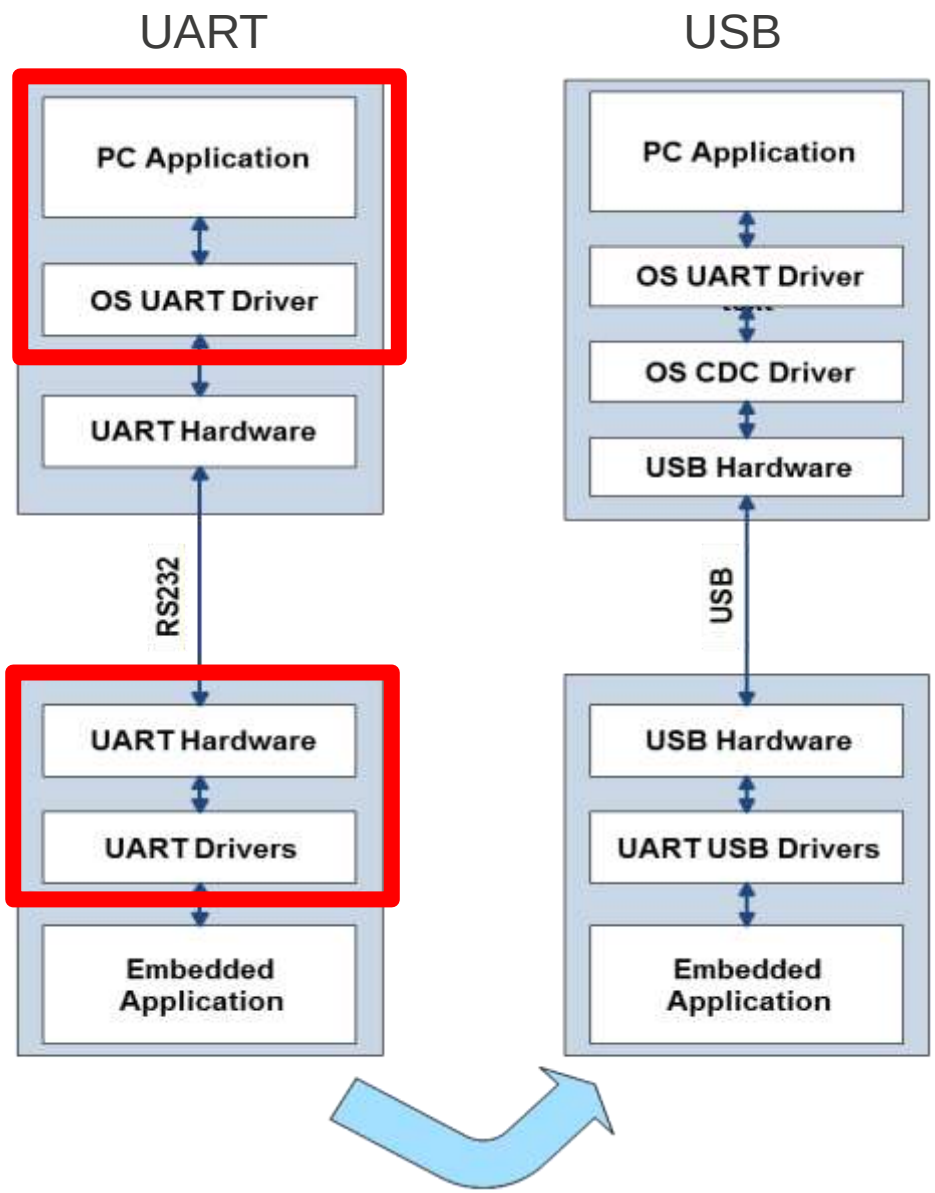
USB CDC Class Overview

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USB CDC Class

Very few modifications needed:

- No change on the PC side: PC sees USB device as a COM port.
- UART driver in the embedded side is replaced with the UART-USB drivers,
- Application calls USB functions instead of UART functions.



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What you need to do

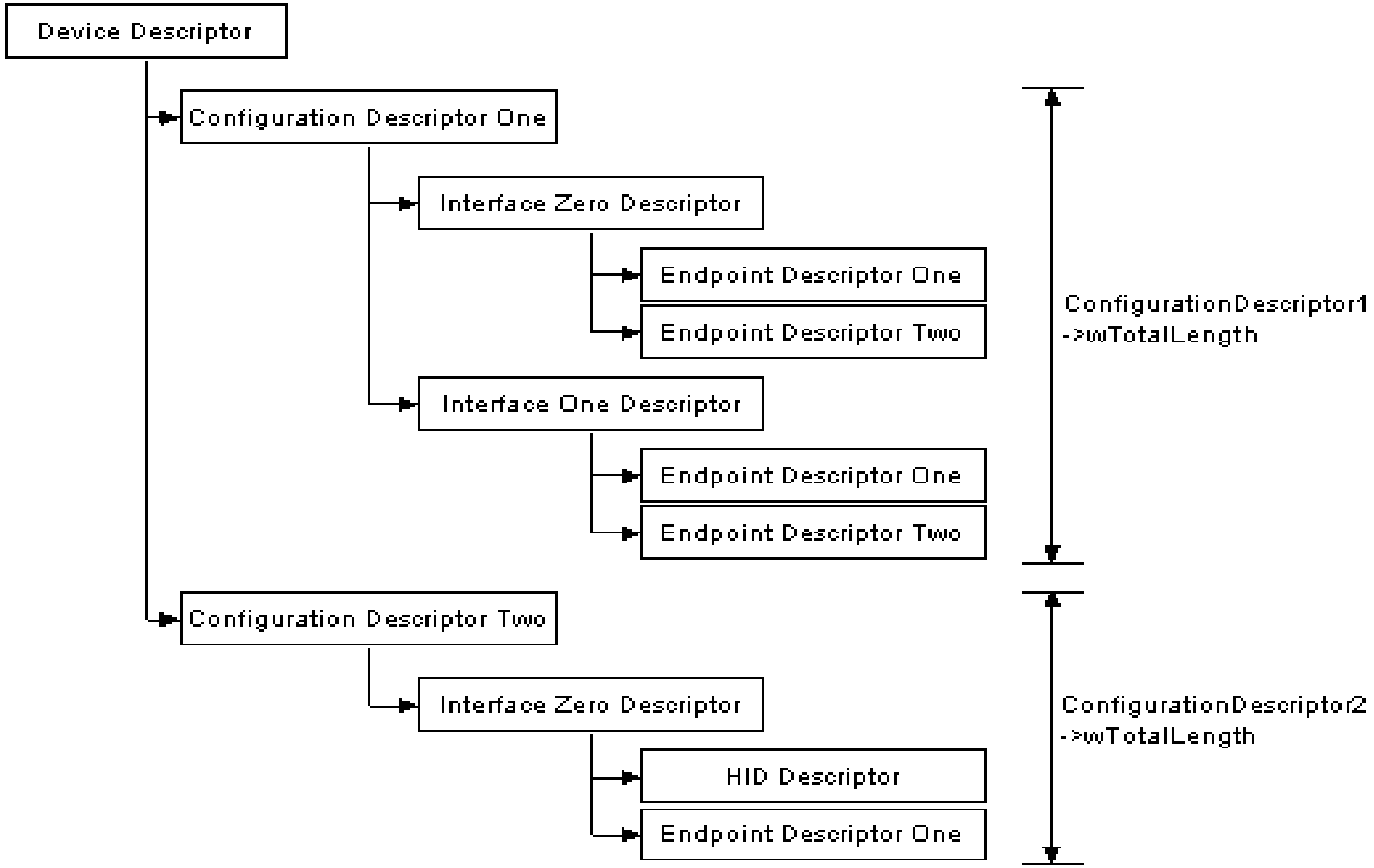
- In your application:
  - Add the Renesas USB FIT module
  - Create or customize the USB ‘descriptor’
  - Initialize USB driver using API calls.
  - Call USB task on periodic basis
    - Your application calls the driver task in a loop or by RTOS task.
  - Send or Receive data using USB API calls.
  - Implement callback functions.

- Install cdc inf driver on the host PC.
  - Customize the .inf file

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Descriptor Hierarchy





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Device Descriptor Example



```
#define USB_VENDORID          (0x0000u)        /* Vendor  ID */
#define USB_PRODUCTID         (0x0002u)        /* Product ID */

/* Standard Device Descriptor */
uint8_t usb_gpcdc_DeviceDescriptor[ 0x12u ] =
{
    0x12u,                                     /* bLength */
    0x01u,                                     /* bDescriptorType (0x01=Device Descriptor Type) */
    0x00u,                                     /* bcdUSB (Lower) */
    0x02u,                                     /* bcdUSB (Upper) */
    0x02u,                                     /* bDeviceClass    (0x02=CDC) */
    0x00u,                                     /* bDeviceSubClass (CDC, unused at this time) */
    0x00u,                                     /* bDeviceProtocol (CDC, unused at this time) */
    0x40u,                                                 /* bMaxPacketSize0 */
    (uint8_t)(USB_VENDORID                    &(uint8_t)0xFFu),  /* idVendor  (Lower) */
    (uint8_t)((uint8_t)(USB_VENDORID >> 8)    &(uint8_t)0xFFu),  /* idVendor  (Upper) */
    (uint8_t)(USB_PRODUCTID                   &(uint8_t)0xFFu),  /* idProduct (Lower) */
    (uint8_t)((uint8_t)(USB_PRODUCTID >> 8)   &(uint8_t)0xFFu),  /* idProduct (Upper) */
    0x00u,                                                   /* bcdDevice (Lower) */
    0x10u,                                                   /* bcdDevice (Upper) */
    0x01u,                                                   /* iManufacturer */
    0x02u,                                                   /* iProduct */
    0x05u,                                                   /* iSerialNumber */
    0x01u                                                    /* bNumConfiguration */
};
```

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The Host Device Class Driver

- Host loads its corresponding driver using PID/VID
- **VID** supplied by USB Implementers forum
  – For fee 

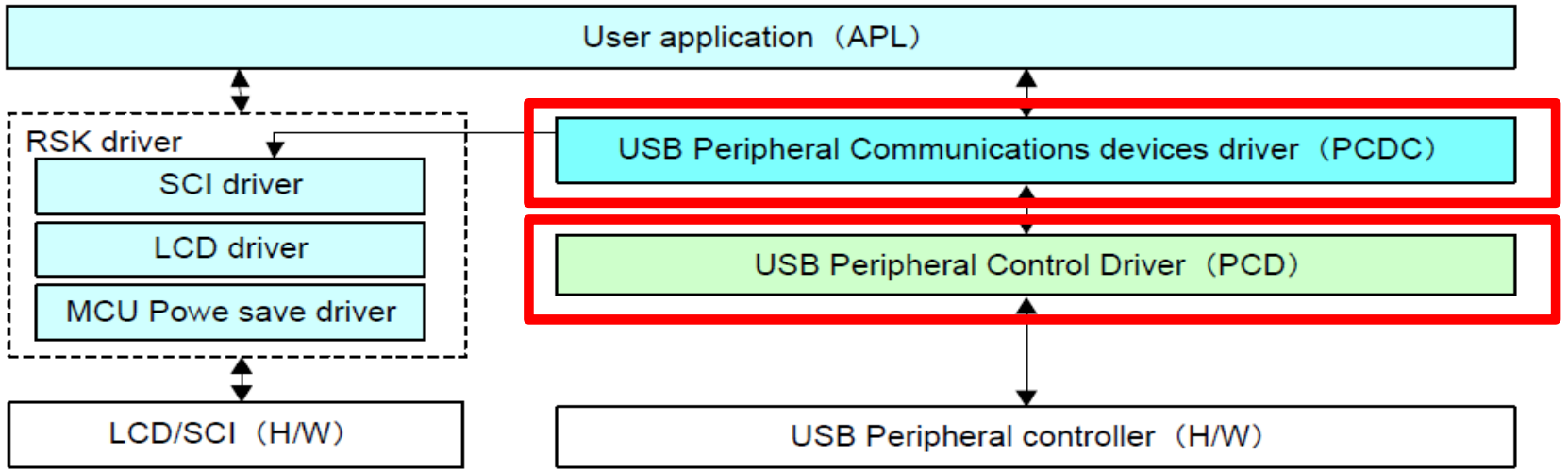
http://www.usb.org

- Host must have .inf file to match DD to load right driver

```
[DeviceList]
%USB\VID_8765&PID_1234.DeviceDesc%=USBBulkInstall, USB\VID_8765&PID_1234
```

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USB Peripheral Communications devices driver PCDC）



Module Name	Function
APL	Sample application program
RSK driver	Sample application for using the peripheral functions of the RSK.
PCDC (r_usb_pcdc)	CDC class driver
PCD (r_usb_basic)	USB peripheral hardware control driver

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USB HID Class Overview

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USB HID Class

The USB human interface device class (USB HID class) consists primarily of devices that are used by humans through physical interaction to control the operation of computer systems. Typical examples of HID class devices include:

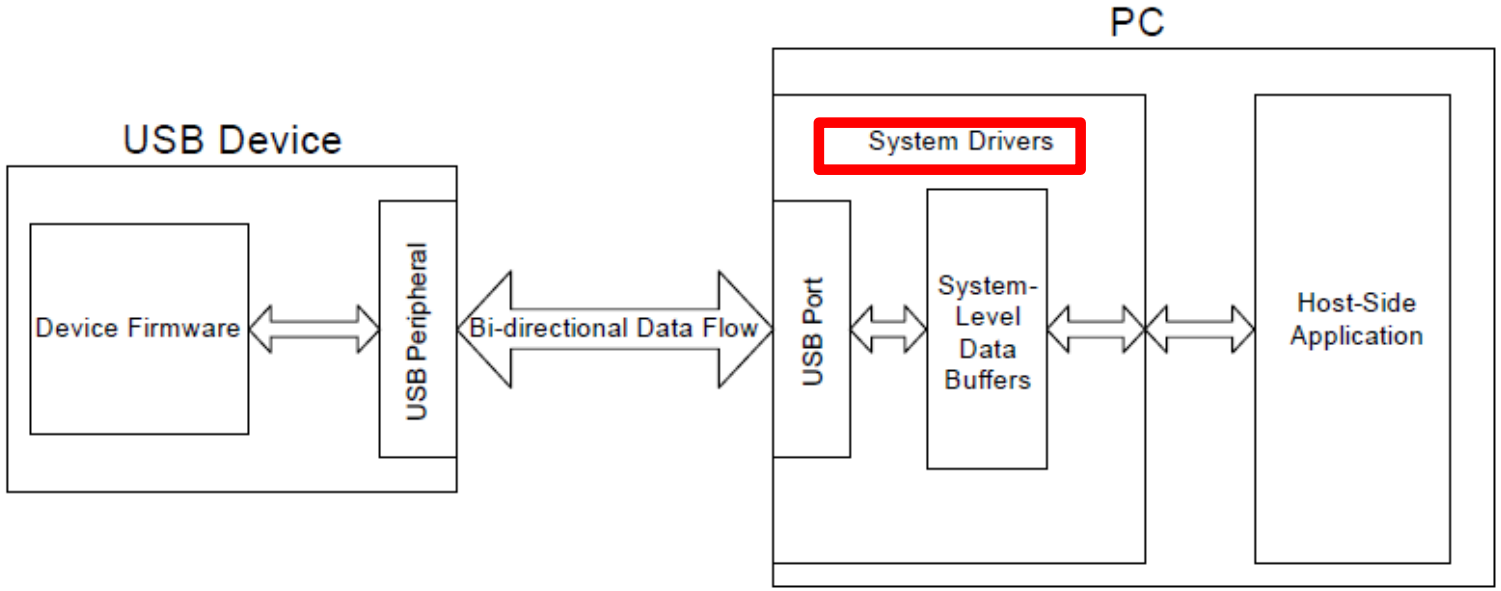
- Keyboards
- Pointing devices: mouse, pen, trackball, joystick.
- Front-panel controls: knobs, switches, buttons, and sliders.
- Touchscreen
- Devices that may not require human interaction but provide data in a similar format to HID class devices—for example, bar-code readers, thermometers, or voltmeters.

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Why HID?

Benefits of developing with HID include:

- Compatibility with all commonly-used computer operating systems
- No need for driver development or distribution
- Streamlined device/host interface via standardized HID specifications



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HID Concepts

Based on a couple of fundamental concepts:

Reports

- The payload data chunk exchanged between a device and a software client.

Report Descriptor

- Describes the format and meaning of each data chunk that it relates to.

  = flexible report format/contents

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HID Reports

HID applications and devices exchange data through Reports.

There are three report types:

- Input Reports
- Output Reports
- Feature Reports

Report Type	Description
Input Report	Data chunks sent from the HID device to the application. Typically when the state of a control changes.
Output Report	Data chunks sent from the host application to the HID device, for example to LEDs on a keyboard.
Feature Report	Data chunks that are manually read or written. Typically related to configuration information

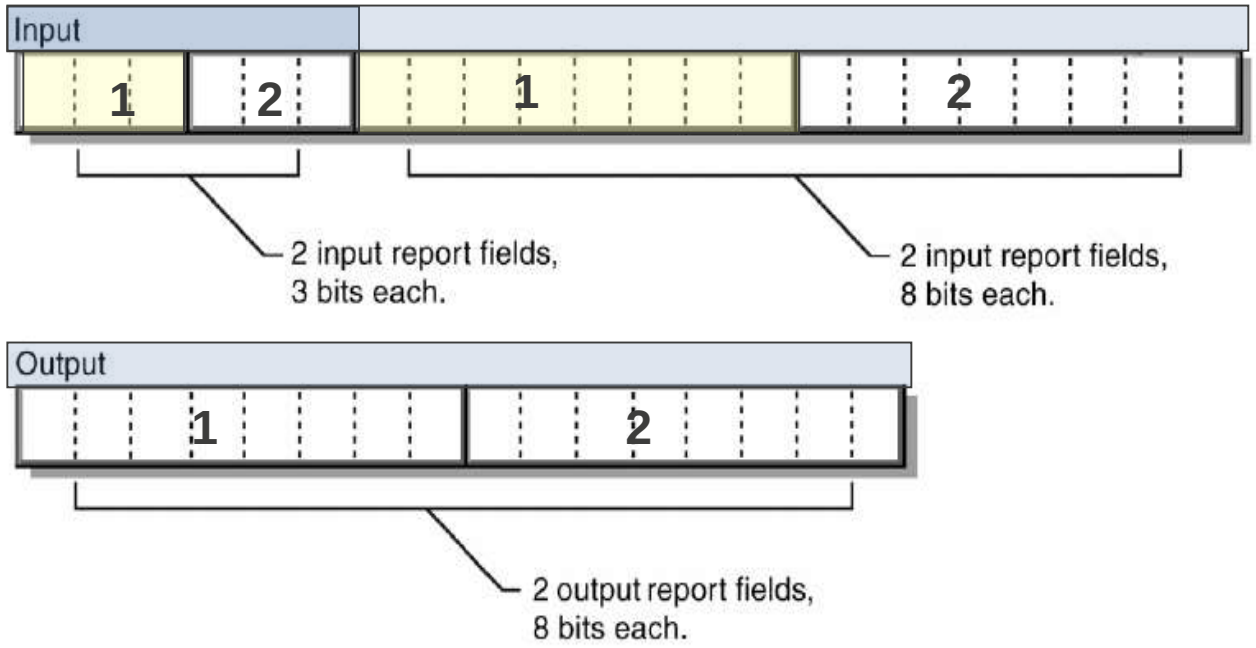
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Report Descriptor and Example (HID)

'Report Descriptor' sent at enumeration

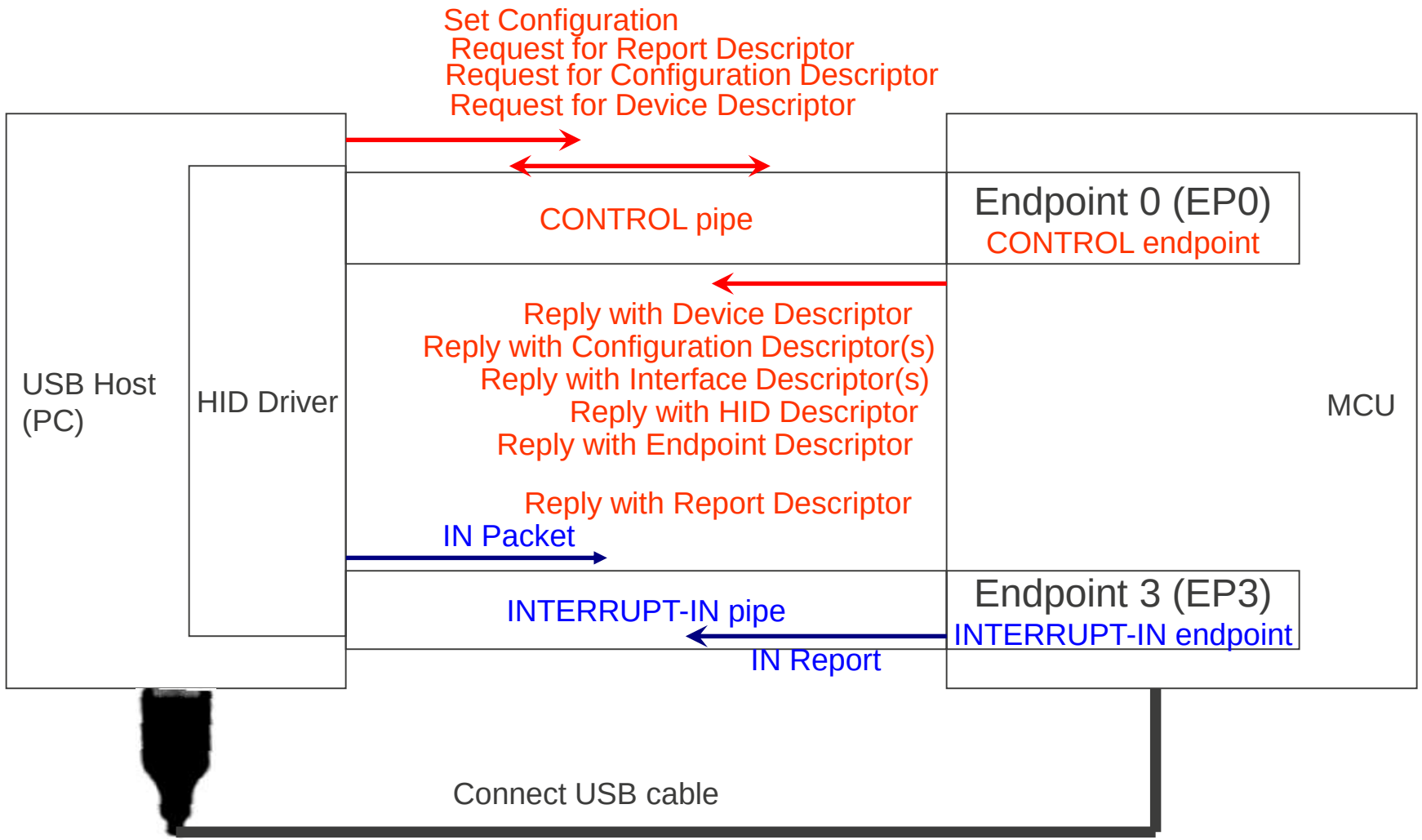
- Explains format of subsequent interrupt data
- Descriptor must follow USB-IF specs

- Example 'HID Report'
- Report Count = 2
- Report Size = 3 bits
- Input
- Change report size to 8
- Input (count still 2)
- Output (count still 2 size still 8)



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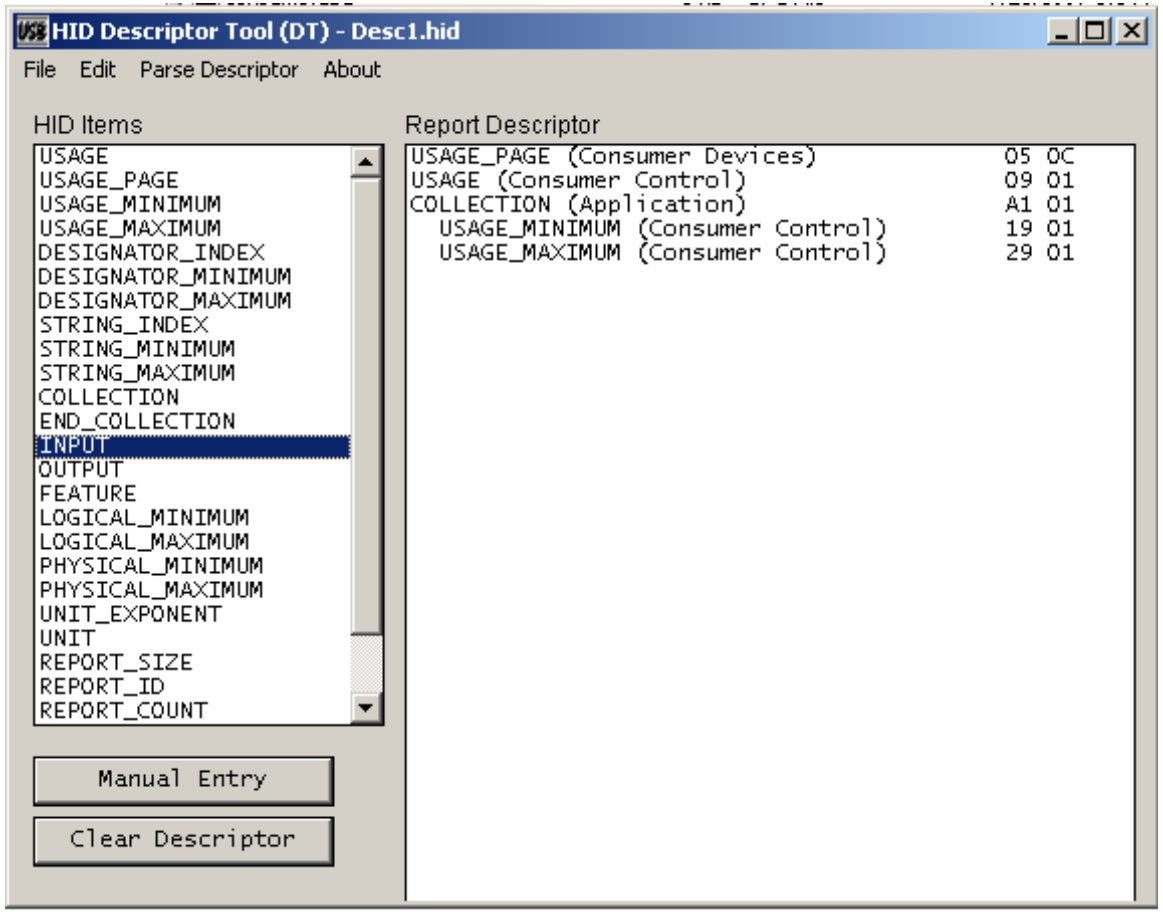
Enumeration Example (HID)



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HID Descriptor Tool (HDT)

- Create, Edit and Validate HID Report Descriptors
- Free from usb.org



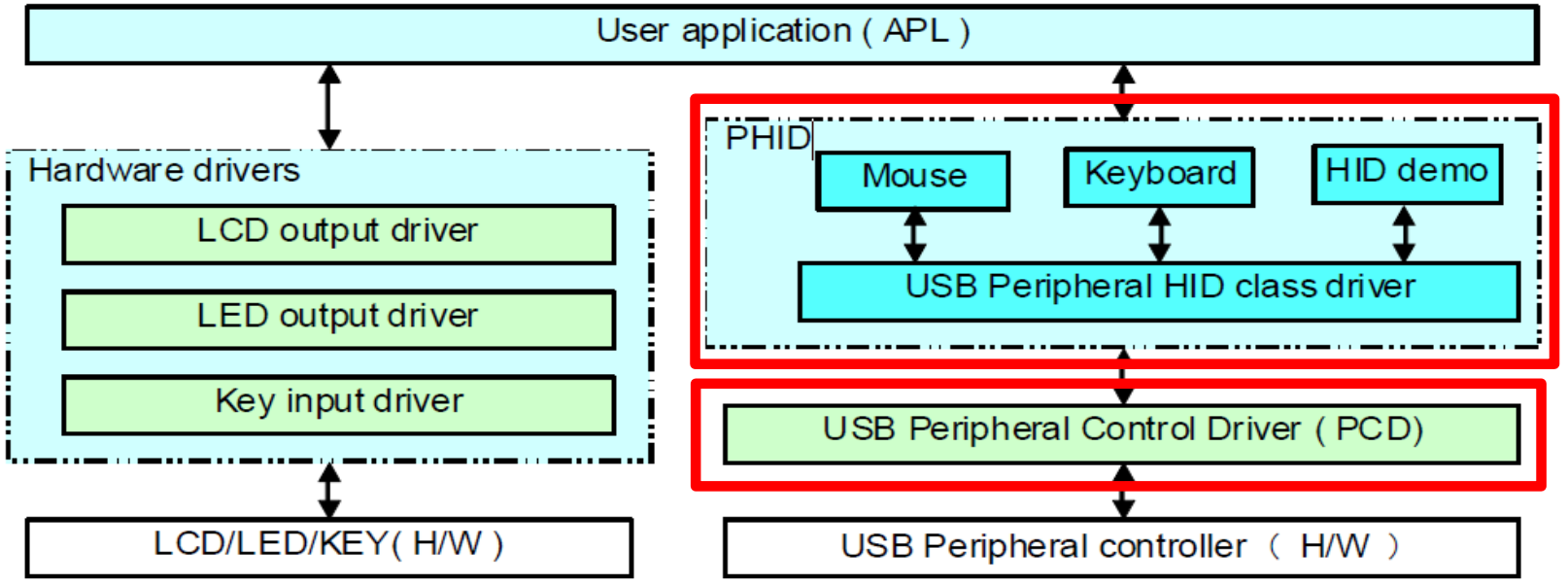
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Renesas USB Peripheral HID class driver

Conforms to the HID Class specification

Has the following main functions:

(1) Response to function references from the HID host
(2) Response to class requests from the HID host
(3) Send Report transfers to USB Host due to user input action on device.



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Hands-on Lab

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Lab Introduction

In the lab exercises you will:

Create brand-new applications and add USB FIT modules.

1. Get a basic understanding of how to apply the Renesas USB peripheral class CDC and HID drivers.
2. Gain insights on methods of using USB for data exchange between PC and an embedded application.

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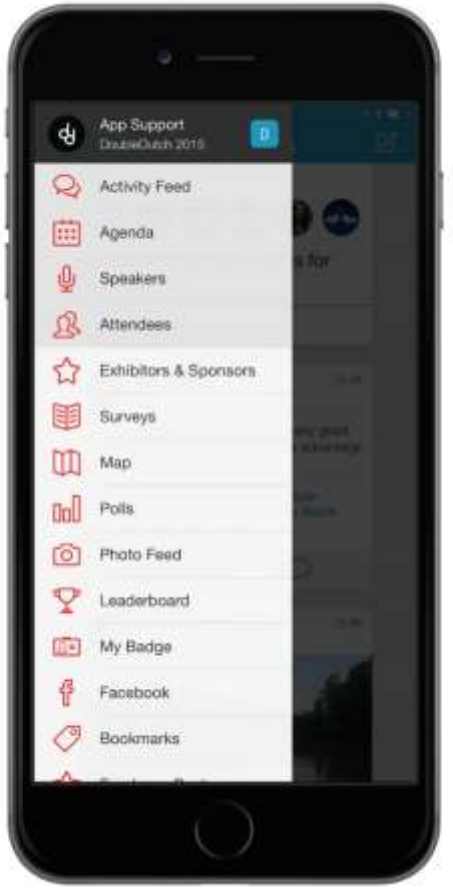
Questions?



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